

RESEARCH/TECHNICAL EXPERIENCES

- **2026, Jan, 6 - 2026, Apr, 4 : Researcher/PhD student (Full-time @ University of Mons)**
 - Developed scientific publications, technical reports, replication packages, and open-source tools, ensuring transparency, reproducibility, and community accessibility.
 - Finalizing doctoral dissertation and preparing for PhD defense.
- **2025, Jan, 5 - 2026, Jan, 5 : Software Engineer/Data Analyst (Full-time @ Necko Technologies)**
 - Engineered a FinOps cost optimization platform leveraging AWS CUR v2 and delivered customer-level cost insights through the CODUS dashboard.
 - Designed and deployed serverless backend services with AWS (Lambda, API Gateway, DynamoDB, RDS, S3), ensuring scalability and high availability.
 - Developed and documented RESTful API architectures, improving integration performance across systems.
 - Optimized relational database schemas and queries to enhance data accessibility and efficiency.
 - Led backend architecture design for cloud-native applications, aligning data models with business objectives.
- **2021, Oct, 21 - 2025, Jan, 4 : Researcher/PhD student (Full-time @ University of Mons)**
 - Collected and processed large-scale datasets from APIs (GitHub REST/GraphQL) and web scraping pipelines.
 - Performed data preprocessing and quality assurance (handling missing data, outlier detection, correlation analysis, k-means clustering).
 - Conducted mixed-method research (quantitative + qualitative), using surveys, interviews, and coding frameworks to extract actionable insights.
 - Produced scientific publications, technical reports, replication packages, and open-source tools, ensuring reproducibility and accessibility.
- **2023, Oct, 10 - 2024, Jan, 3: Visiting Researcher @ Radboud University**
 - Designed and conducted mixed-method studies, combining quantitative analyses with qualitative approaches such as surveys, interviews, and coding frameworks to generate empirically grounded insights.
 - Worked under the co-supervision of Mairieli Wessel, deepening expertise in qualitative research methodologies, which resulted in a peer-reviewed publication as the outcome of the research stay.
- **2019, Oct, 1 - 2021, Sept, 20: Research Assistant/Master Student (Part-time @ ISE Lab @ SUT)**
 - Automated the linking of commits to issue reports in large-scale GitHub repositories by designing data collection pipelines and constructing research-grade datasets.
 - Applied and evaluated machine learning models (e.g., XGBoost, Decision Trees, Random Forest) to improve the accuracy of issue-commit traceability.
 - Conducted empirical software engineering research, implemented experiments, and authored scientific publications documenting findings and methodologies.
- **2019, Jan, 1 - 2019, Aug, 1: Research Assistant/Software Engineer (Internship @ IIPL @ KNTU)**
 - Built ETL pipelines to integrate data from public datasets, GitHub APIs, and web scrapers for research use.
 - Enhanced dataset reliability by imputing missing values, removing anomalies, and applying statistical validation.
 - Delivered analytical reports, visualizations, and dashboards, enabling researchers to derive data-driven insights.

TEACHING EXPERIENCES

- **Feb, 2 - May, 24 of 2022, 2023, 2024: Software Evolution** To computer science **master** students @ Faculty of Sciences @ UMONS (Dates vary a bit due to academic year calendar)
 - Designed and developed hands-on TP (lab) materials aligned with course objectives and semester planning
 - Contributed to course organization, including scheduling, assessment design, and scoring guidelines
 - Evaluated students based on TP performance, project deliverables, and final presentations
- **2020, Sept, 1 - 2020, Dec, 10: Data Science for Software Engineering** To Software Engineering **master** students @ Faculty of Sciences @ SUT
 - Prepared and graded homework assignments contributing to students' final evaluations
 - Led lab and discussion sessions to deepen students' understanding of core data science concepts
- **2019, Jan, 10 - 2019, May, 20: Database Design** To Software Engineering **bachelor** students @ Faculty of Sciences @ SUT
 - Developed and graded homework assignments forming part of students' final scores
 - Conducted lab and discussion sessions to reinforce database theory and practical skills
- **2018, Sept, 1 - 2018, Dec, 10: Data Structure & Database Design** (two separate courses) To Software Engineering **bachelor** students @ Faculty of Sciences @ KNTU
 - Created and evaluated homework assignments across both courses
 - Delivered lab sessions and guided students through problem-solving and practical exercises

EDUCATION

University of Mons (UMONS)	Mons, Belgium
Ph.D. in Software Engineering, Advisor: Prof. Tom Mens	2021, Oct, 21 – Current
– Thesis title: “Understanding the Evolution of GitHub Actions Automation Workflows”	
Sharif University of Technology (SUT)	Tehran, Iran
M.S. in Software Engineering, Advisor: Dr. Abbas Heydarnoori, GPA: 18.30/20	2019, Sept, 1 – 2021, Sept, 20
– Thesis title: “Automatic Linking of Issue Reports and Commits in Software Repositories”	
K. N. Toosi University of Technology (KNTU)	Tehran, Iran
B.S. in Computer Engineering, Advisor: Dr. Ali Ahmadi, GPA: 17.64/20	2015, Sept, 1 – 2019, Jun, 20
– Thesis title: “Deaf Disease Genes Detection Using Artificial Intelligence Tools”	

EMPLOYMENT

University of Mons (UMONS)	Mons, Belgium
Bourse de Doctorant, Ph.D. in Software Engineering, Advisor: Prof. Tom Mens	2026, Jan, 6 – 2026, Apr, 5
Necko Technologies	Mons, Belgium
Cloud Engineer	2025, Jan, 5 – 2026, Jan, 5
University of Mons (UMONS)	Mons, Belgium
Bourse de Doctorant, Ph.D. in Software Engineering, Advisor: Prof. Tom Mens	2021, Oct, 21 – 2025, Jan, 4

PUBLICATIONS (GOOGLE SCHOLAR LINK)

1. **P. Rostami**, A Decan, T Mens, “gawd: A Differencing Tool for GitHub Actions Workflows”, *Mining Software Repositories conference(MSR)*, 2024, Tool's link: <https://github.com/pooya-rostami/gawd>
<https://doi.org/10.1145/3643991.3644873>
2. **P. Rostami**, A Decan, T Mens, M Wessel, “A preliminary study of GitHub Actions workflow changes”, *Seminar on Advanced Techniques and Tools for Software Evolution (SATToSE)*, 2023,
<https://ceur-ws.org/Vol-3483/paper8.pdf>

3. M Wessel, T Mens, A Decan, **P. Rostami**, “The GitHub Development Workflow Automation Ecosystems”, *Software Ecosystems: Tooling and Analytics, Book chapter, Springer Nature, 2023*,
https://doi.org/10.1007/978-3-031-36060-2_8
4. **P. Rostami**, T Mens, A Decan, M Golzadeh, “On the usage, co-usage and migration of CI/CD tools: a qualitative analysis”, *Empirical Software Engineering Journal (EMSE)*, 2023,
<https://doi.org/10.1007/s10664-022-10285-5>
5. A Decan, T Mens, **P. Rostami**, M Golzadeh, “On the Use of GitHub Actions in Software Development Repositories”, *International Conference on Software Maintenance and Evolution(ICSME)*, 2022,
<https://doi.org/10.1109/ICSME55016.2022.00029>
6. N chidambaram, **P. Rostami**, “Bot Detection in GitHub Repositories”, *Mining Software Repositories conference(MSR)*, 2022,
<https://doi.org/10.1145/3524842.3528520>
7. **P. Rostami**, M Izadi and A. Heydarnoori, “Automated Recovery of Issue-Commit Links Leveraging Both Textual and Non-textual Data”, *International Conference on Software Maintenance and Evolution(ICSME)*, 2021,
<https://doi.org/10.1109/ICSME52107.2021.00030>

OPEN SOURCE TOOLS

- **gawd (GitHub Actions Workflow Differ) – Python** 2024
Developed an open-source Python library and CLI tool to compute structural differences between GitHub Actions workflows, identifying added, removed, moved, renamed, or modified items with syntax-aware comparison.

SCHOLARSHIPS AND AWARDS

- ICSME 2022 Distinguished Paper Award for “On the Use of GitHub Actions in Software Development Repositories”. (<https://cyprusconferences.org/icsme2022/>)
- MSR 2022 best Hackathon Paper Award for “Bot detection in GitHub repositories”. (<https://conf.researchr.org/info/msr-2022/awards>)

VOLUNTEERING

- **Junior PC @ International Conference on Technical Debt (TechDebt 2026)** Nov 2025 - Jan 2026
Serving as a reviewer for two papers (one industry experience paper and one research paper) on technical debt for the TechDebt track, co-located with ICSE 2026. More info: <https://conf.researchr.org/track/TechDebt-2026/TechDebt-2026-junior-program-committee>
- **Workshop publicity chair for BENEVOL 2022** Apr 2022 - Sept 2022
Creating and maintaining of BENEVOL 2022 workshop website (<https://benevol2022.github.io/>) plus Helping with arranging the workshop.
- **Presented qualitative research methodology as a volunteer PhD student** Mar 2023
Delivered a full-session lecture to master’s students on conducting qualitative studies as part of PhD Month activities.
- **Participating in peer review**
Reviewer for journal papers submitted to the Journal of Systems and Software (JSS).

TECHNICAL SKILLS

- **Software Development Technologies & Methodologies:** Agile (Scrum), DevOps, IDEs (VS Code, IntelliJ, JupyterLab), testing framework (pytest), Git, Atlassian (Bitbucket, Jira), Docker, Linux/Unix
- **Coding:** Python, TypeScript, HTML, CSS
- **Database:** RDBMS such as MySQL, MS SQL
- **Data Analysis Libraries & Technologies:** Pandas, Keras, sci-kit learn, NumPy, NLTK, HuggingFace, PySpark
- **ML Algorithms:** Classical Machine learning, and deep learning algorithms (SVM, Random Forest, CNN, LLMs)
- **AWS:** S3, Glue, Athena, Lambda, Step Functions, RDS, Kinesis, QuickSight, etc.

LANGUAGES

English: Fluent

French: Intermediate Proficiency (B1 ongoing)

Persian: Native